



INDIAN INSTITUTE OF INFORMATION TECHNOLOGY,
DESIGN AND MANUFACTURING,
KANCHEEPURAM



Robotics

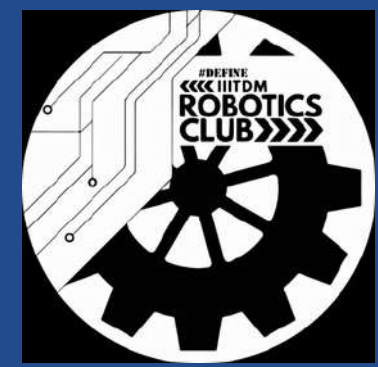
*MOTOR CONTROL AND SENSOR
INTEGRATION*

ROBOTICS CLUB



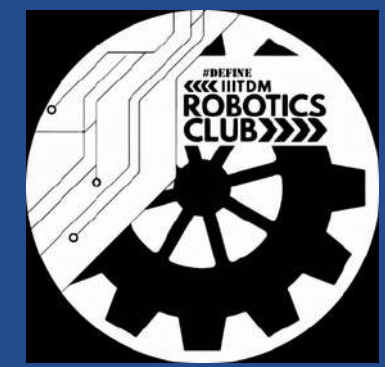


INDIAN INSTITUTE OF INFORMATION TECHNOLOGY,
DESIGN AND MANUFACTURING,
KANCHEEPURAM



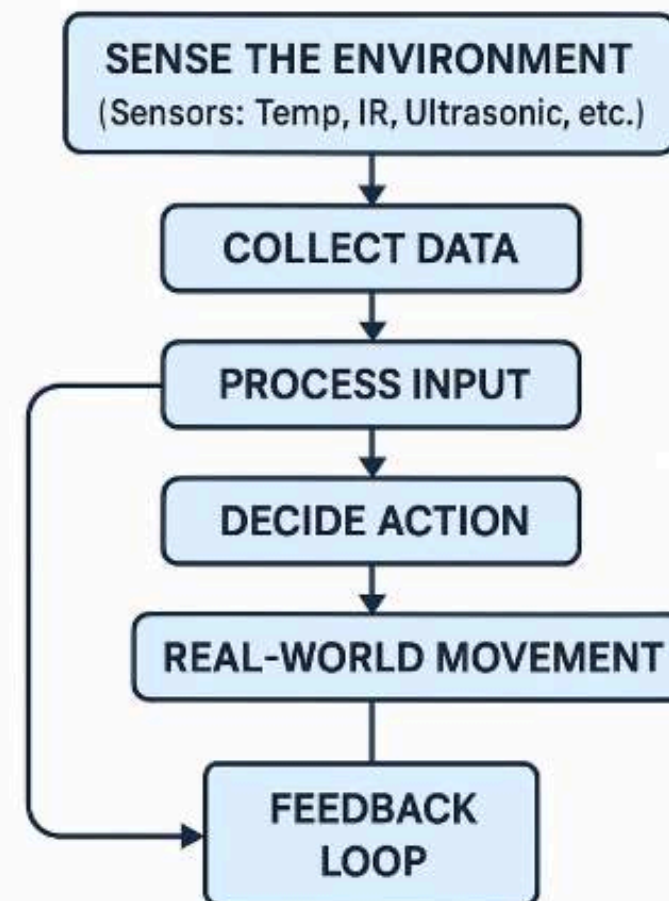
Prerequisites

Your presence



MOTIVATION?

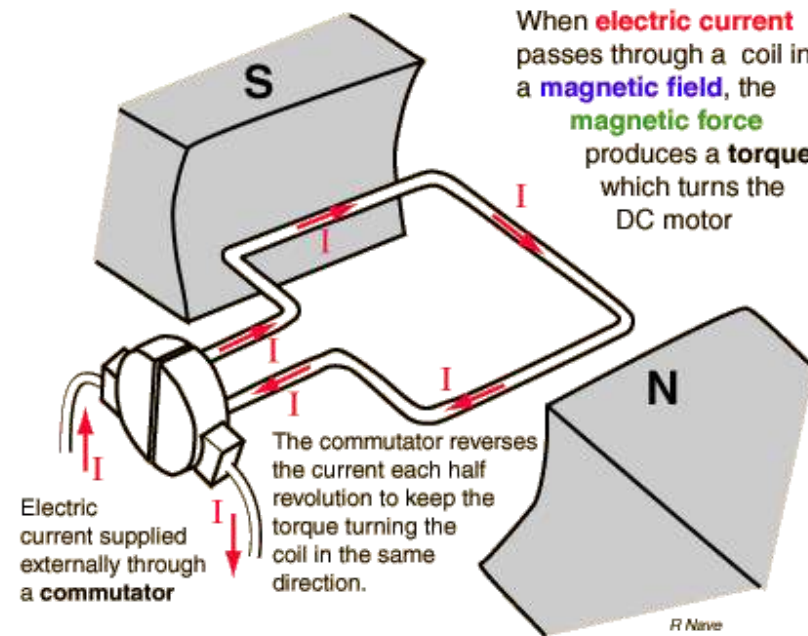
WHY LEARN MOTOR CONTROL & SENSOR INTEGRATION?



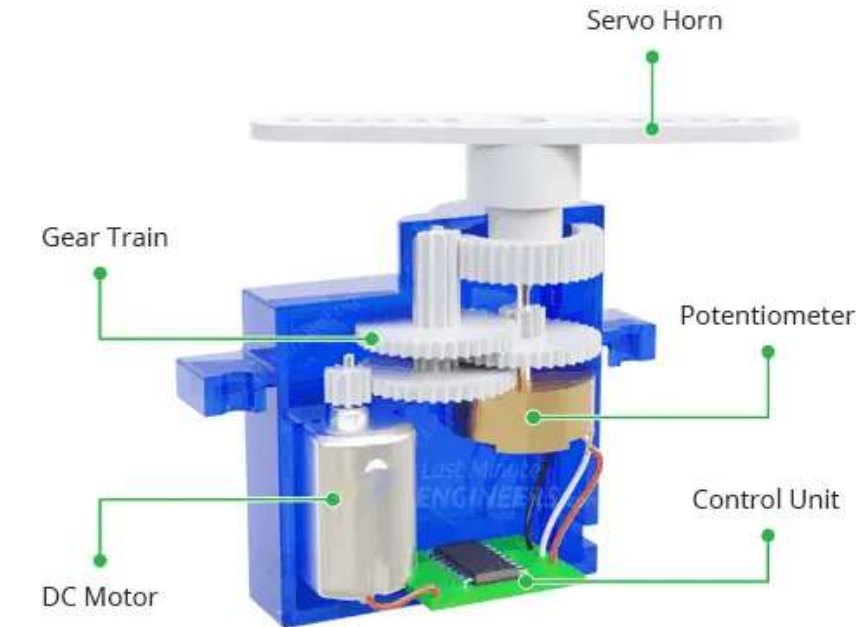
- ✓ Enables smart systems
- ✓ Core of robotics & automation
- ✓ Used in drones, robots, appliances, vehicles

RECAP

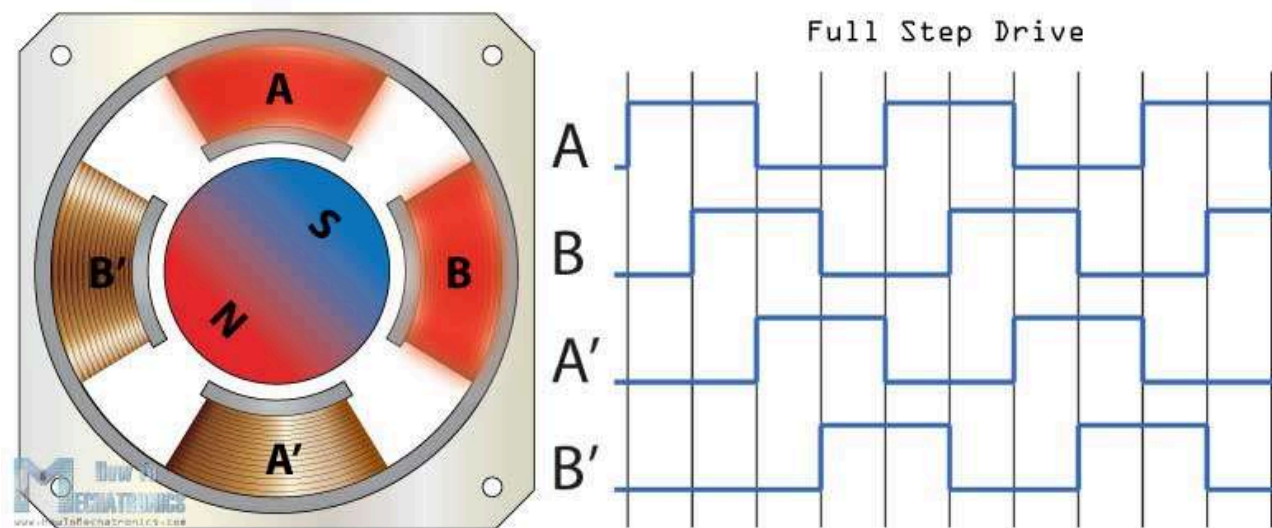
DC MOTOR



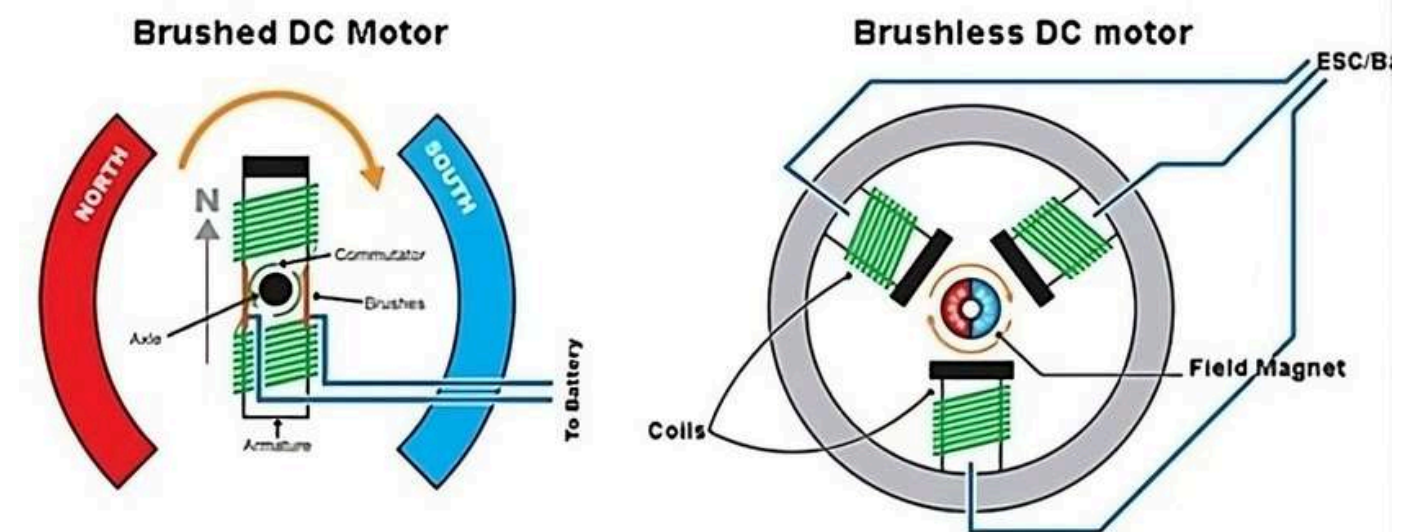
SERVO MOTOR



STEPPER MOTOR

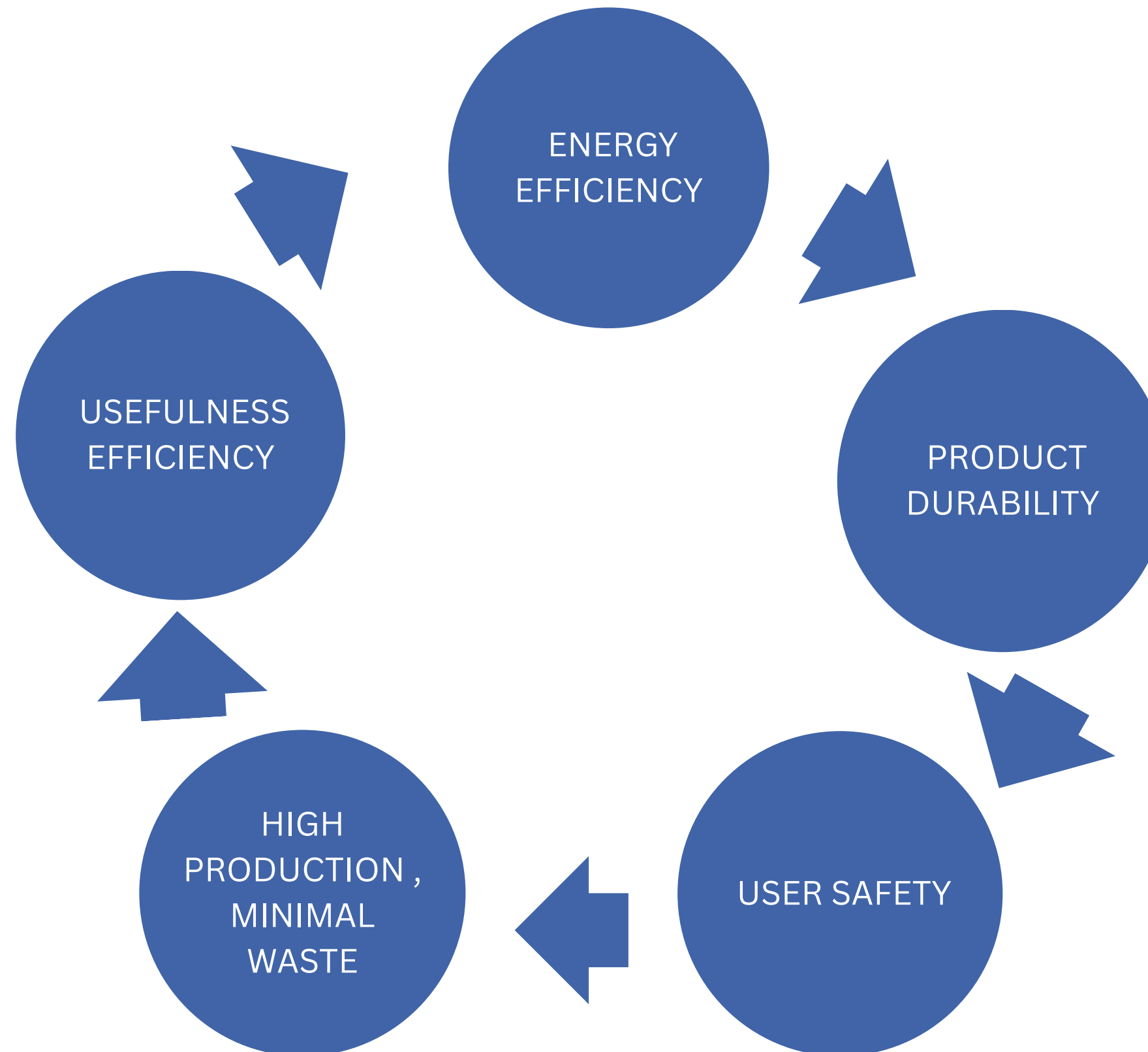


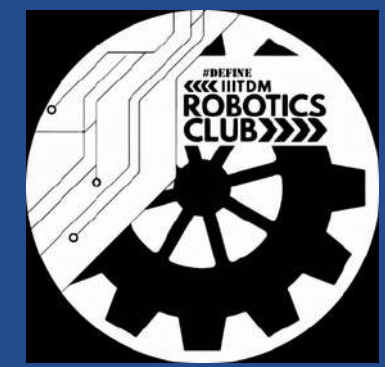
BLDC MOTOR





NEED :-





MAIN COMPONENTS OF A MOTOR CONTROL CIRCUIT :-

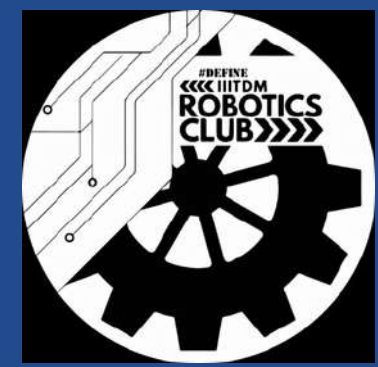
Motor (DC, Stepper, Servo, or BLDC)

Driver Circuit (e.g., H-Bridge)

Power Supply

Sensors (e.g., encoders, current sensors)

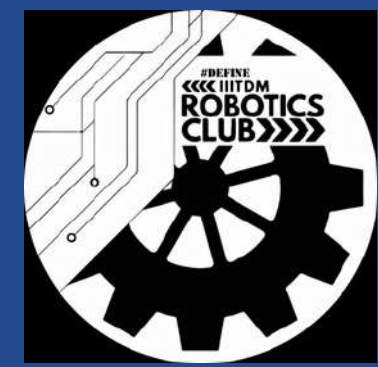
Controller (MCU, DSP, PLC)



CONTROL STRATEGIES

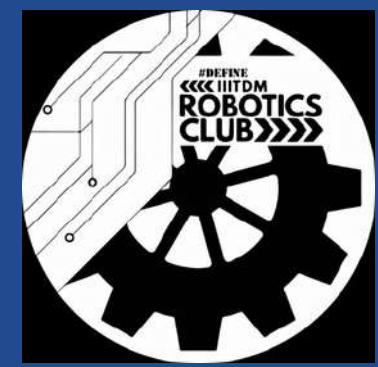
OPEN-LOOP CONTROL:

CLOSED-LOOP CONTROL:



KEY CIRCUIT TECHNIQUES :-

- **Pulse-Width Modulation (PWM)**
- **H-Bridge for direction control**
- **Current sensing for protection and torque control**
- **Feedback using position/speed sensors**



MOTOR TYPES AND THEIR CONTROL METHODS :-

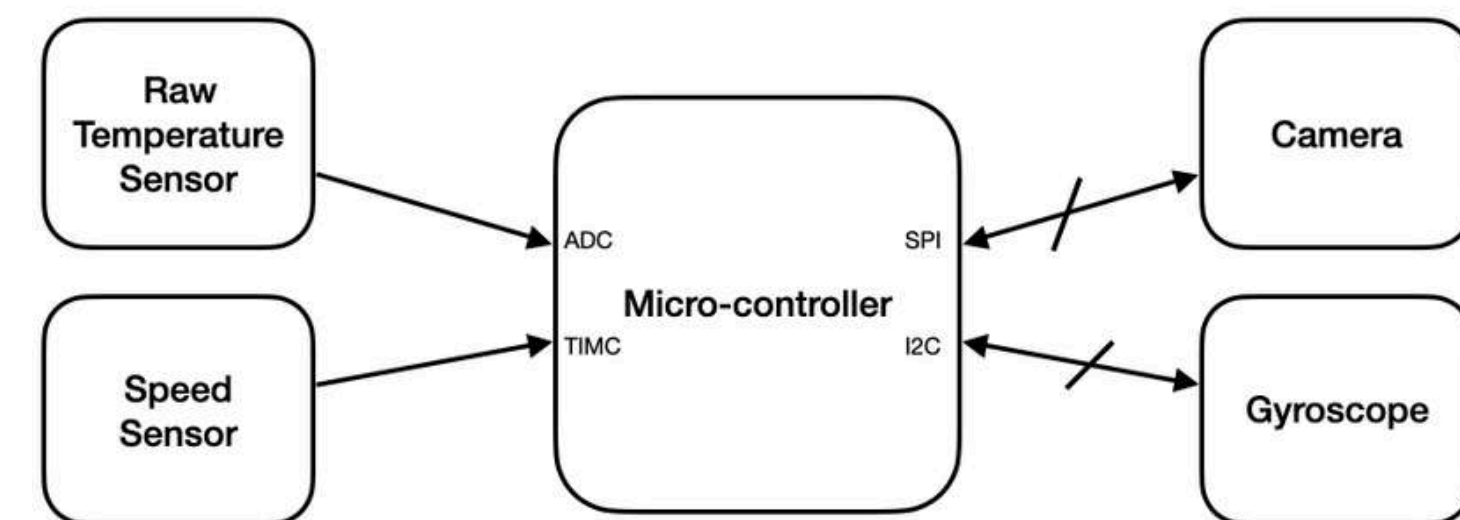
- DC Motor: PWM for speed, H-Bridge for direction**
- Servo Motor: PWM for position control**
- Stepper Motor: Pulse train for step-based motion**
- BLDC Motor: Electronic commutation with sensors**



SENSOR INTEGRATION :-

A key aspect of robotics that allows machines to sense and interact with their environment. Sensors act as the eyes, ears, and touch of a robot, collecting data like distance, light, or motion, which the robot uses to make decisions.

A sensor detects physical properties like distance, temperature, or pressure and converts them into electrical signals. These signals are sent to a microcontroller, such as an Arduino or Raspberry Pi, which processes the data using code to control motors, lights, or other components. Integration involves hardware connections (wiring the sensor correctly) and software (programming the microcontroller to interpret sensor data).



Key Steps in Integration:

- **Selection:** Choose a sensor based on the task (e.g., detecting obstacles or measuring temperature).
- **Connection:** Wire the sensor to the microcontroller, ensuring proper power and signal lines.
- **Programming:** Write code to read sensor data and define how the robot responds.
- **Calibration:** Adjust the sensor to ensure accurate readings in the specific environment.

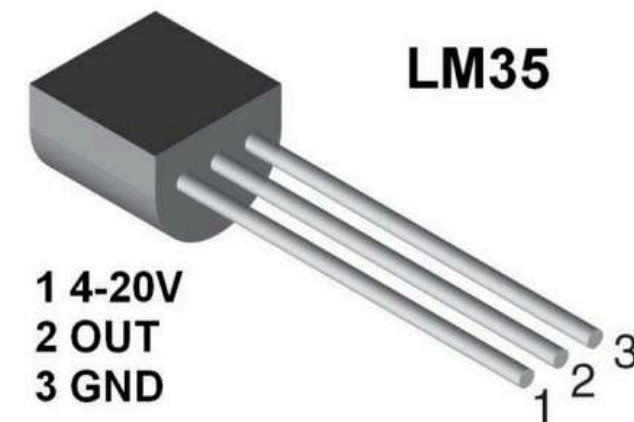
Common Sensors used in Projects :-



Ultrasonic Sensor



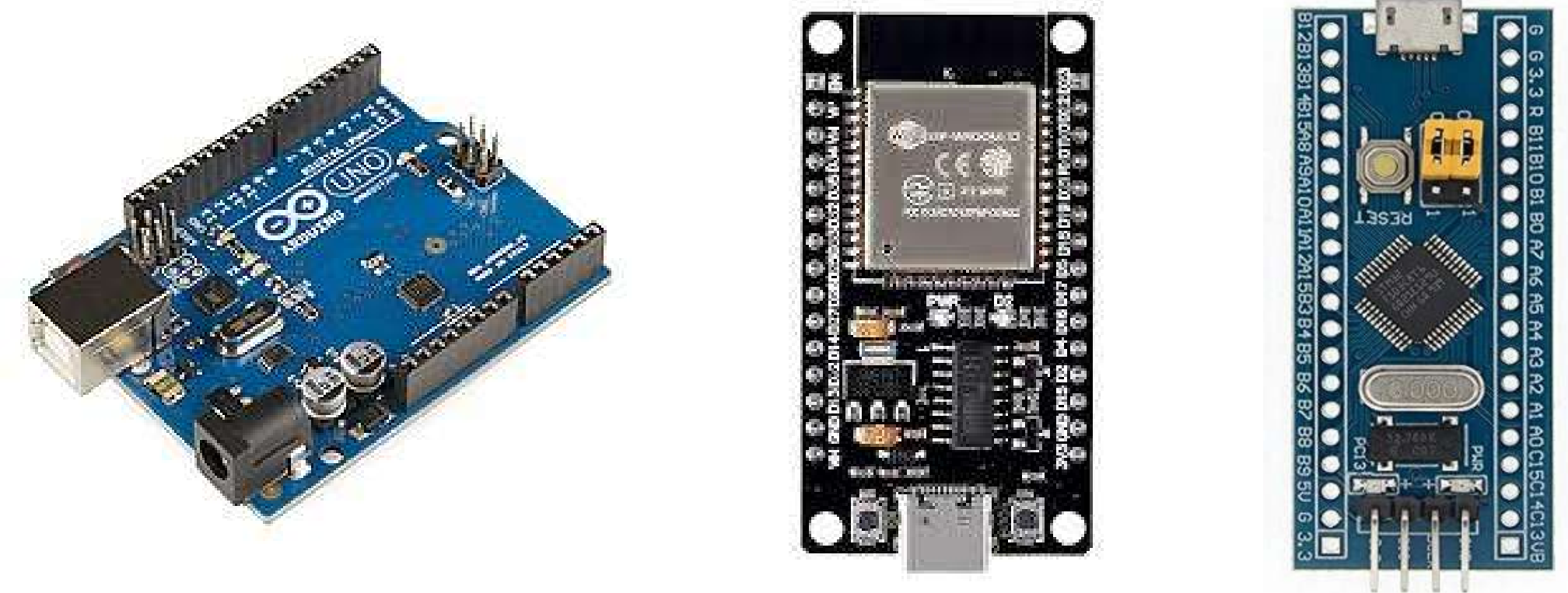
IR Sensor



Temperature Sensor



Smoke Sensor



Pins which convert analog inputs to digital inputs (ADC Converter)

For analog sensors, the microcontroller's ADC converts the voltage to a digital value, which software maps to the measured property (e.g., voltage to temperature). For digital sensors, the microcontroller reads pulse durations, high/low states, or protocol data. Libraries, like the Wire library for I2C



INDIAN INSTITUTE OF INFORMATION TECHNOLOGY,
DESIGN AND MANUFACTURING,
KANCHEEPURAM



QnA



Resources

All about Motor Control?

How Stuff Works

Complete Guide on Sensor Integration :-

MIT

All about Motors

electronoobs

A Beginner's Guide to Choosing Motors

core electronics

QUT Robot Academy



Connect with us



@ROBOTICSCLUB.IITDM

**INSTAGRAM
PAGE**



**ROBOTICS CLUB
LINKEDIN PAGE**



**IGNITE
WHATSAPP GROUP**



**TECH AFFAIRS
LINKTREE**





Thank You



Share your feedback about this session